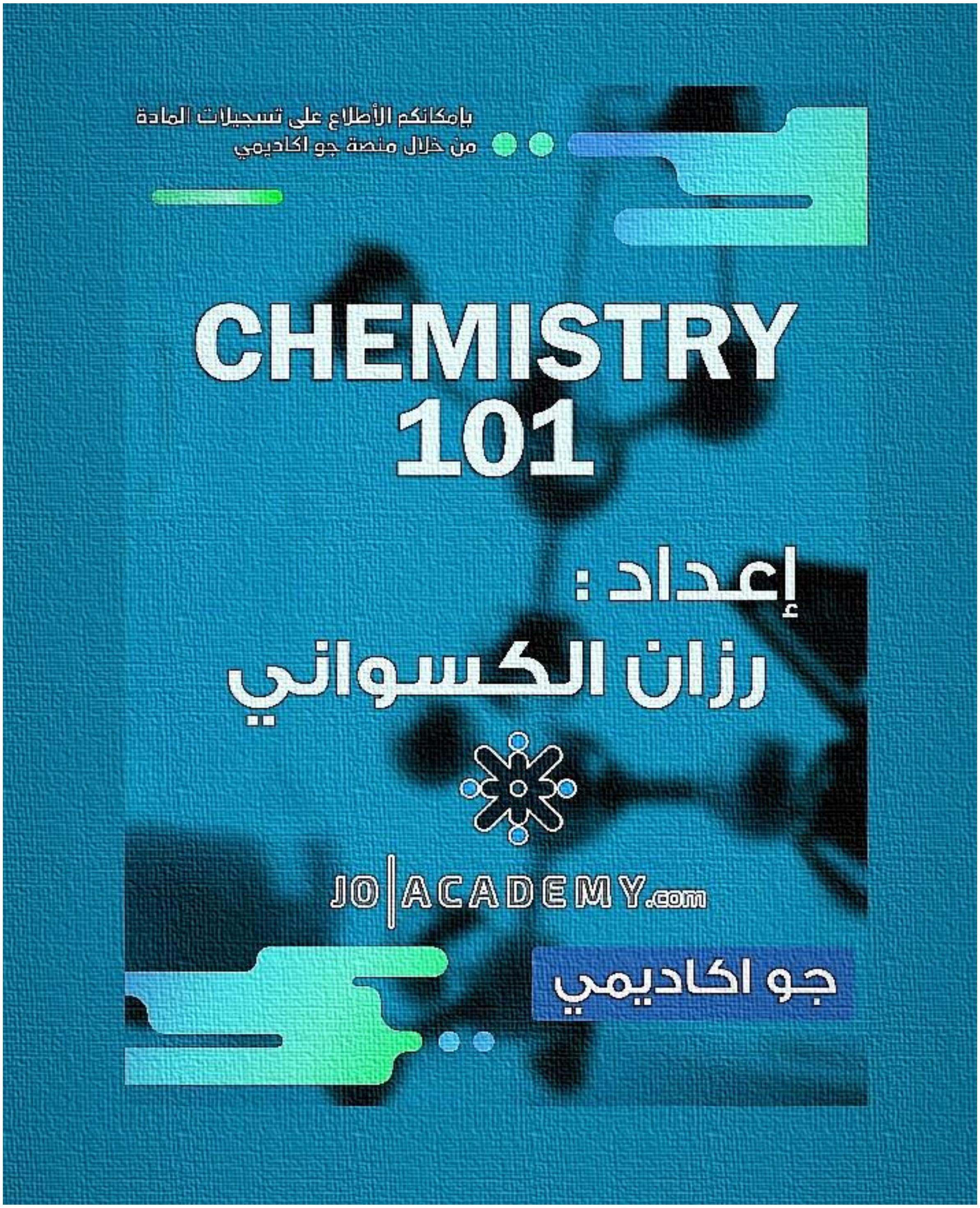
بإمكانكم الأطلاع على تسجيلات المادة
من خلال منصة جو اكاديمي
CHEMISTRY
101
إعداد :
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After carrying out the operation below, how many significant figures are appropriate to show in the result?

(13.7 + 0.027) / 8.221

(13.7 +0.027) = 13.727 = 13.7

(13.7 / 8.221) = 1.664

1.66 = 3 Significant Figures

How many orbitals are allowed in a subshell if the angular momentum quantum number for electron in that subshell =3?

L = 3 → F orbital

So, # of subshell orbitals = 7 عدد الافلاك الفرعية

Calculate the frequency of light emitted by hydrogen atom during transition of its electron from (n = 4 to n = 1):

$$\frac{1}{\boldsymbol{\lambda}} = 1.097 * 10^{7}\left(\frac{1}{ni2} - \frac{1}{nf^{2}}\right) = 1.097 * 10^{7}\left(\frac{1}{16} - 1\right) = -10284375$$

$$\frac{1}{\boldsymbol{\lambda}} = \mathbf{10284375} \ldots \ldots \ldots \ldots\ \boldsymbol{\lambda} = \mathbf{9.723} * \mathbf{10^{-8}}\ \boldsymbol{m}$$

$$v = \frac{C}{\boldsymbol{\lambda}} = \frac{\mathbf{3} * \mathbf{10^{8}}}{(\mathbf{9.723} * \mathbf{10^{-8}})} = \mathbf{0.3085} * \mathbf{10^{16}}\ \boldsymbol{Hz}$$

How many sulfur atom are there in 21 g of (Al2S3):

$$\#\text{of atoms} = \#\text{of mole} * \text{Avogadro's number} * \text{S atom number}$$

- Molar Mass of (Al2S3) = 150.158 g/mole

$$\#\text{ of S atoms} = \left(\frac{21}{150{,}158}\right) * 6.02 * 10^{23} * 3 = 2.525 * 10^{23}\text{ atoms}$$

Calculate the molarity (mole/Liter) of an aqueous solution of ammonia that has 5.5 % NH3 and 125 ml (volume), its density is 0.788 g/ml:

- Mass of NH3 = 5.5 g

$$Density = \frac{mass}{volume} ===> Mass = Density * Volume$$

$$Mass = 0.788\frac{g}{ml} * 125\ ml = 98.5\ g\ NH3$$

Consider the following reaction ($Cu + \frac{1}{2} O_2 \rightarrow CuO$), if 2.35 g of Cu reacted with 2.35 g of O_2, then the theoretical yield is:

- Molar Mass of Cu = 63.546 g/mole
- Molar Mass of O_2 = 32 g/mole

Cu	O_2
2.35 g	2.35 g
Mole = mass / molar mass = 2.35 / 63.546 =0.03698 mole	Mole = Mass /Molar Mass =2.35 / 32 =1.175 mole
0.03698 / 1 = 0.03698	1.175 / (1/2) = 2.35
اقل عدد مولات Cu	
Limiting Reactant = Cu	

1 mole Cu → 1 mole CuO
0.03698 Mole Cu →?? Mole CuO

Mole of CuO = 0.03698 mole
Molar mass of CuO = 63.546 + 16 = 79.546 g/mole
Mass of CuO = 0.03698 * 79.546 = 2.9416 g

The electron configuration of carbon atom:

C (6): $1S^2\ 2S^2\ 2P^2$

The combustion of 0.465 g of sample (CxHyOz) yield 0.66 g of CO2 and 0.405 g of H2O.what is the empirical formula of this sample:

- $CO_2 \rightarrow$ 1 atom C (m.m of CO_2 = 44 g/mole)
- $H_2O \rightarrow$ 2 atom H (m.m of H_2O = 18 g/mole)
- $Mole\ of\ C = \#\ of\ C\ atom * \left(\frac{mass\ of\ CO2}{Molar\ mass\ CO2}\right) = 1 * \left(\frac{0.66}{44}\right) = 0.015\ mole\ C$
- $Mass\ of\ C\ atom = Mole\ of\ C * Molar\ Mass\ of\ C$
- $Mass\ of\ C = 0.015 * 12 = 0.18\ g$
- $Mole\ of\ H = \#\ of\ H\ atoms * \left(\frac{mass\ of\ H2O}{molar\ mass\ of\ H2O}\right) = 2 * \left(\frac{0.405}{18}\right) = 0.045\ mole\ H$
- $Mass\ of\ H\ atom = mole\ of\ H\ atom * molar\ mass\ of\ H\ atom$

  $Mass\ of\ H\ atom = 0.045 * 1 = 0.045\ \ g$
- $Total\ Mass\ of\ sample = H\ mass + C\ mass + O\ mass$

  $$0.465 = 0.045 + 0.18 + mass\ of\ O$$
- Mass of O atom = 0.24 g

	C	H	O
Mass	0.18 g	0.045	0.24
Mole	(0.18 / 12) =0.015	(0.045 / 1)= 0.045	(0.24 / 16) = 0.015
اقل عدد مولات 0.015			
Atom	0.015/0.015 = 1	0.045 / 0.015 = 3	0.015 / 0.015 = 1
Sample = CH3O			

The electron configurations of (Gd) atom (64 electron) is:

Gd (64): $1S^2\ 2S^2\ 2P^6\ 3S^2\ 3P^6\ 4S^2\ 3d^{10}\ 4P^6\ 5S^2\ 4d^{10}\ 5P^6\ 6S^2\ 4F^7$

Electronic Configuration (Using Xe element = 54 electron)

Gd = [Xe] $6S^2\ 4F^7$

In one process, 6 Kg of CaF_2 are reacted with an excess of (H_2SO_4) and yield 2.86 Kg of HF. Calculate the percent yield of HF ($CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$)

- Molar Mass of CaF_2 = 78.07 g
- Molar Mass of HF = 20.01 g/mole

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- Mole of CaF_2 = Mass / Molar Mass = 6000 /78.07 = 76.85 mole

- 1 mole $CaF_2 \rightarrow$ 2 mole HF

  76.85 mole $CaF_2 \rightarrow$?? Mole HF

  - Mole of HF = 153.7 mole

- Mass of HF = Mole * Molar Mass = 153.7 * 20.01 = 3075.5 g

- $\%\ Yield = \frac{Actual}{theoritical} * 100\% = \frac{2860}{3075.5} * 100\% = 92.99\ \%$

What is the combination of protons, neutrons and electrons is correct for the isotopes of copper (Cu^{63})

Atomic number of Cu = 29   من الجدول الدوري

# of proton = 29

# of electron = 29

# of neutron = Mass number - # of proton = 63-29 = 34